

Network Layer:

The network layer is the third layer in the OSI (Open Systems Interconnection) model. It is responsible for the logical addressing, routing, and forwarding of packets across network boundaries. The primary objective of the network layer is to determine the best path to transfer data from the source to the destination across multiple networks.

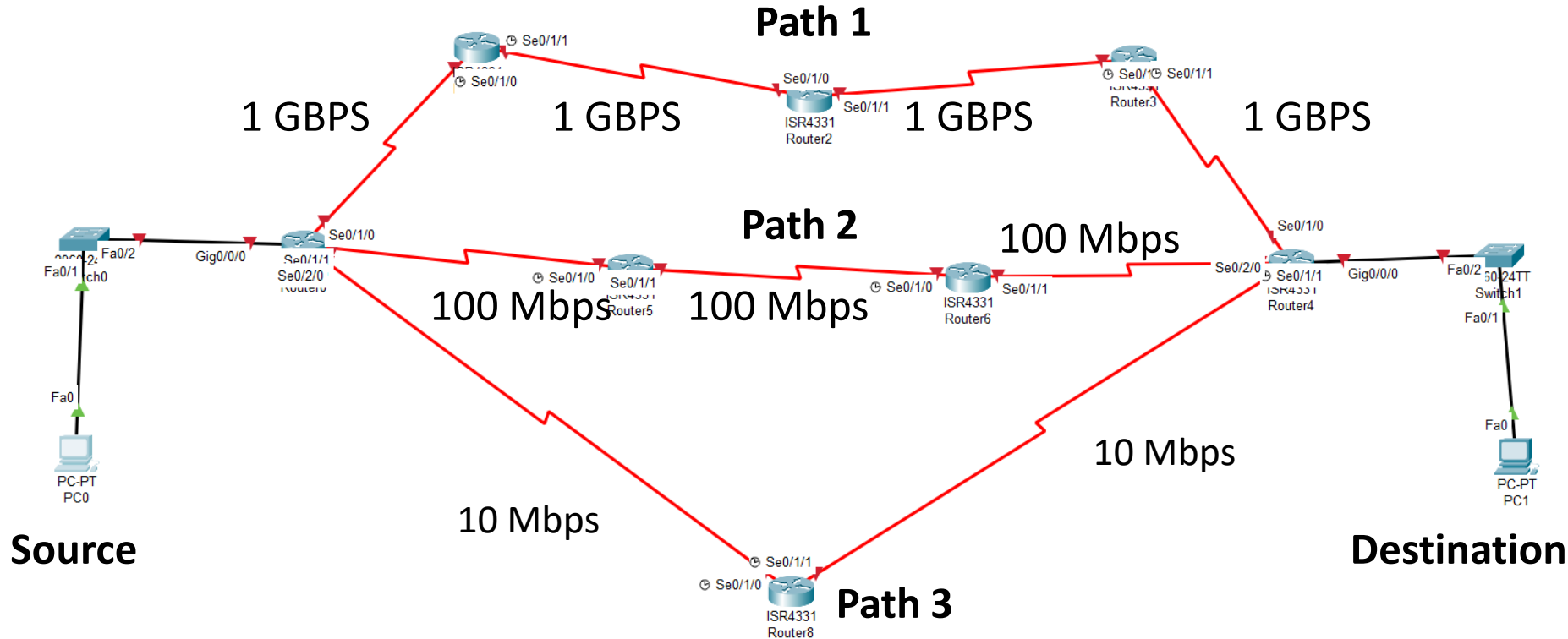
Key Functions of the Network Layer

- **Logical Addressing**
- **Routing**
- **Packet Forwarding**
- **Packet Fragmentation and Reassembly**
- **Error Handling and Diagnostics**

Logical Addressing: The network layer provides logical addressing (e.g., IP addresses) that devices use to identify themselves on a network. Unlike physical (MAC) addresses, which are unique to network interfaces, logical addresses can be assigned and managed dynamically.

Routing: One of the core functions of the network layer is routing. Routing involves determining the optimal path for data packets to travel from the source to the destination. Routers, which operate at the network layer, use routing tables and algorithms to make these decisions.

Best Path Selection: Which one is best Route?



Packet Forwarding: Once the path is determined, the network layer forwards the packets from one network segment to another, eventually reaching the destination. This involves handling the packets at each hop along the route.

Packet Fragmentation and Reassembly: If a packet is too large to be transmitted over a network segment, the network layer can fragment the packet into smaller units. These fragments are reassembled back into the original packet at the destination network layer.

Error Handling and Diagnostics: The network layer is responsible for error handling and diagnostics. Protocols like ICMP (Internet Control Message Protocol) are used to report errors and provide network diagnostic functions (e.g., ping and traceroute).

Protocols at the Network Layer:

- Internet Protocol (IP)
- ICMP (Internet Control Message Protocol)
- Routing Protocols
 - . OSPF (Open Shortest Path First)
 - . BGP (Border Gateway Protocol)
 - . RIP (Routing Information Protocol)

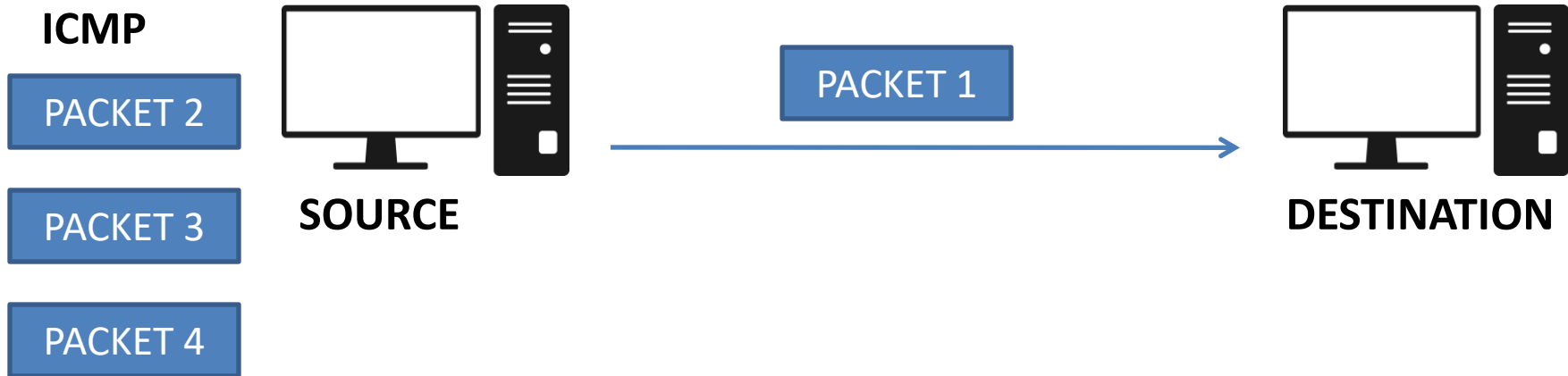
IP Address: The most widely used network layer protocol. IP provides logical addressing and routing functions. There are two versions in widespread use:

-IPv4 (Internet Protocol version 4): The older and more widely deployed version, which uses 32-bit addresses.

-IPv6 (Internet Protocol version 6): The newer version designed to address the limitations of IPv4, using 128-bit addresses to provide a much larger address space.

ICMP (Internet Control Message Protocol):

Used for error reporting and diagnostic functions. ICMP messages are used by network devices to send error messages indicating that a service is not available or that a host or router could not be reached.



Packet Size: 32 Bytes and forwards 4 Packets to confirm the packet Forwarding